QSCP Corporate Asset Integrity

Corporate Summary

No.	Status	Date
QSCP-CI-SUM-2023-04	Issued	18.09.2023

Corporate Integrity Summary - Anlage 21

Inspection Results and Compliance

The periodic ZÜS inspection documented in ANG-DA-2023-04018 for ANLAGE-21 confirmed the integrity of the piping systems via external examination and Festigkeitsprüfung. No deficiencies were noted during the formal assessment, and the existing Prüfkonzept remains valid following the 2019 crack finding on NR-21-L-0415. However, the report issued a specific Hinweis regarding wall thickness degradation on line NR-21-L-0207.

Author Daniel Hartmann from QSCP Corporate Asset Integrity notes that while the ZÜS inspection passed, subsequent localized measurements have identified a nonconformity requiring corporate oversight.

Wall Thickness Nonconformity on NR-21-L-0207

Line NR-21-L-0207 (P235GH, Do 88.9 mm × e_n 4.0 mm) transports condensate from E-2103 to B-2107 with design limits of PS 6 bar(g) and TS 165 °C. Per NR-AB-2023-04, a wall thickness violation was recorded at Messstelle 05. Grid measurements detailed in RLW-PB-2023-0201 identified a minimum thickness of 1.7 mm over a 40 × 60 mm area on the elbow.

This value is below the Werksnorm t-min of 2.0 mm. Historical CML data for NR-21-L-0207-05 shows a steady decline from 4.00 mm in 2003 to 1.89 mm in March 2023 and 1.85 mm in July 2023. The calculated remaining life (RL) has expired, with the corrosion rate most recently measured at 0.106 mm/y.

Sign-off	Name	Function	Date	Signature
Prepared	Daniel Hartmann	Integrity Engineer, QSCP Corporate Asset Integrity, based at Werk Niederrhein	18.09.2023	_____